

Saakshi — working prototype

A resilient, provable CBT exam ecosystem: every answer is signed and hash-chained at the seat, the system survives relay, cell and link failures, and any tampering afterwards is locatable and recoverable.

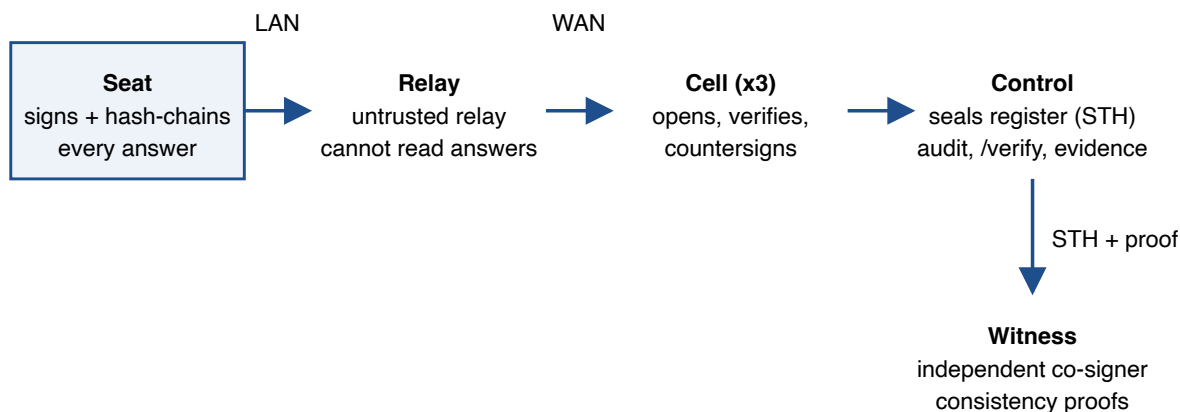
Repository: <https://github.com/anirudh12032008/saakshi>

Tested on macOS and Windows (Windows: automated CI on windows-latest, incl. end-to-end test of the packaged seat)

This document shows that Saakshi is running software, not a concept. Five scripted demo acts drive real seat, relay, cell and control processes (not mocks) and each writes an evidence file to `docs/evidence/`. The design never auto-penalises a candidate: flags go to a human reviewer.

Key limit, stated up front: a seat signature proves a record was not altered after the device produced it, not that the candidate chose it. All analytics figures use synthetic data and all performance figures are loopback on one host.

Architecture



- **Seat** — Electron app at the exam PC; signs and hash-chains locally, fsyncs, works offline.
- **Relay** — untrusted store-and-forward at the centre; sees only sealed envelopes.
- **Cell** — one of three independent national-DC processes; verifies chains, countersigns, seals a per-shift Merkle register.
- **Control** — signs the tree head, runs the audit, serves `/verify` and the evidence pack.
- **Witness** — separate process that cosigns a tree head only with a valid RFC 9162 consistency proof; a rewritten register raises a P0 TAMPER incident. In the demo it shares a machine with control; independent custody is roadmap.

What runs today

Act	What it proves	Measured number	Evidence file
1 Before	Integrity gate blocks a seat running AnyDesk and an overlay tool by name; re-check turns green; face flag goes to human review	2 tools named and blocked; 1 face flag cleared by reviewer	stage5-act1.txt , stage5-gate-selftest.json
2 T0	2-of-3 custody release; centre with cut link unlocks via phoned offline code; forged relay key rejected	308 candidates, 7 centres, 6,586 entries committed	stage8-act2.txt
3 During	WAN cut, cell wipe and rebuild, seat force-quit and move; candidates keep answering	RPO 0; RTO 1.3 to 1.9 s across three drill runs	stage4-chaos.jsonl , stage4-act3.txt
4 After	Rogue DB edit is located; /verify shows "Q17: record says C, the seat committed B" and recovers B; evidence pack manifest verifies	1,000 of 1,000 byte-flip locations found	stage0-tamper-lab.txt
5 Decide	Analytics on the cells' own export flag likely cheating and decide compensate, re-test or re-conduct	203,031 entries; 53 flags; no honest candidate flagged; headline: Compensated 0, Re-tested 193, Re-conducted 3 centres, Spared 1,190 (synthetic)	stage6-act5.txt , stage6-act5-full.txt

Measured numbers

One Apple M5 Mac (16 GiB), 3 real cells on loopback, 99 simulated centres, synthetic G1 cohort. Not real WAN.

Measure	Result	Source (docs/evidence/)
Ingest latency, relay to cell (3,697 calls)	p50 44.1 ms, p99 168.1 ms	stage7-load.txt
Entries stored / rejected at 20k candidates	227,882 / 0	stage7-load.txt
Candidates unlocked	19,803 of 19,803	stage7-load.txt
Relay-to-cell traffic	about 1.06 MB per candidate-hour	stage7-load.txt
Seat CPU / RSS	2.3% median (max 4.3%) / 166 MiB	stage7-load.txt
Chaos drill (kill-cell, wipe-cell, spare-relay)	20/20 pass, 0 of 16,000 answers lost, RTO median 0.96 to 1.23 s	stage8-chaos.txt
Offline-start act	PASS, 308 candidates at 7 centres, 6,586 entries	stage8-act2.txt

Caveats: the per-candidate-hour figure extrapolates a 60 s window; seat CPU/RSS cover the Node main process over about 10 s; everything is loopback on one host.

The running system

Stills from the recorded demo (submitted separately). Burned-in text is the video's narration caption.

Control · DEMO-2026 · S1

Clean seats only

CEN042

1 ready · 1 amber · 0 review · 1 blocked · 5 not reported
Blocked

1. C0001 · CEN042-S01 · Blocked — AnyDesk

2. C0003 · CEN042-S03 · Amber — battery

Review queue

Decide as

e.g. REVIEWER-1

Act 1. Control board: seat blocked by name (AnyDesk), another amber on battery; review queue shows a cleared face flag.

0 ready · 0 amber · 0 review · 0 blocked · 8 not reported
Ready

Review queue

Decide as

e.g. REVIEWER-1

Start on time

Candidates registered

308

Seats bound

301

Paper unlocked

301

Submitted

281

Entries per second

40

6 of 7 centres green · 3 of 3 exam servers live

cell-1: live · 1,971 entries · cell-2: live · 2,154 entries · cell-3: live · 2,257 entries

Act 2. Custody release: 6 of 7 centres green, Centre 42 (link cut) still locked; 308 candidates registered.

P3 notice

C0001's old seat is still sending

1 centre · 1 candidate · answers lost 0

open 0:24 · [invigilator] · superintendent · control · regulator

stays with the invigilator

its entries after the move are kept as evidence (ORPHANED), not counted

Acknowledge ORPHANED-9

Resolve ORPHANED-9

P1 major

Data Centre 2 is rebuilding from the relays

1 cell · 2 centres · 100 candidates · answers lost 0

closed after 0:10 · invigilator · superintendent · [control] · regulator

closed

0 of 2 relays have replayed

Act 3. Incident cards during chaos: Data Centre 2 rebuilding from the relays, "answers lost 0".

Choose fileNo file chosen

Receipt code from the slip (optional)

V4Q9627Z3QK42DXTY

Loaded the sealed register (C0002): candidate C0002, DEMO-2026 S1, 13 entries.

Witness: not cosigned

The record was altered after the seat committed it

Q17: record says C — the seat committed B

Check	Result	Detail
Seat keys (certified by the exam server, or pinned DEV keys)	✓ pass	1 key epoch(s) for C0002: pinned DEV keys
Signed hash chain	✓ pass	13 entries: signatures, sequence and links verify

Act 4. /verify page: "Q17: record says C — the seat committed B"; signed chain and receipt checks shown.

Decide fairly

Incident id

Breach perimeter

Evidence

Leaked items

☐ Systemic

Run

e.g. act5-small

comma-separated centres, o

how the breach was four

optional, comma-se

Compensated 0 · Re-tested 193 · Re-conducted 3 centres · Spared 1,190 · ₹ avoided 17,85,000

C00156 · CEN010

Watch

speed-accuracy

correct on 19 of 19 hard items answered unusually fast (ability decile 1; expected 3.3)

corroboration only — no prior attempt in the registry: nothing to corroborate

C00243 · CEN010

Watch

speed-accuracy

Act 5. Decision headline on the cells' own export: Compensated 0, Re-tested 193, Re-conducted 3 centres, Spared 1,190 (synthetic).

Measured numbers	
One Apple M5 Mac (16 GiB), 3 real cells on loopback, 99 simulated centres, synthetic G1 cohort. Not real WAN.	
Ingest latency, relay to cell (3,697 calls)	p50 44.1 ms, p99 168.1 ms
Entries stored / rejected at 20k candidates	227,882 / 0
Candidates unlocked	19,803 of 19,803
Relay-to-cell traffic	about 1.06 MB per candidate-hour
Seat CPU / RSS	2.3% median (max 4.3%) / 166 MiB
Chaos drill (kill-cell, wipe-cell, spare-relay)	20/20 pass, 0 of 16,000 answers lost, RTO median 0.96 to 1.23 s
Offline-start act	PASS 308 candidates at 7 centres, 6 586 entries committed

Close. Measured-numbers slide: every figure is cited to a file in the evidence folder.

Engineering proof

Check	Result
Workspace tests (<code>pnpm -r test</code>)	476 passing: core 93, seat 98, server 285
Analytics tests (pytest)	76 passed (<code>docs/evidence/stage6-pytest.txt</code>)
Continuous integration	Green on macOS and Windows. Windows CI on windows-latest: full workspace tests, packaged-seat build and probe self-test, gate self-test (renamed AnyDesk.exe and overlay tool blocked by name), check that the build refuses <code>--inspect</code> and <code>--remote-debugging-port</code> , and a Playwright end-to-end test of the packaged seat. A separate server job runs on each commit.
Chaos drill	20/20 runs pass; 16,000 sent = 16,000 stored, 0 answers lost (<code>stage8-chaos.txt</code>)
Tamper lab	1,000 of 1,000 byte-flips located (<code>stage0-tamper-lab.txt</code>)
Protocol spec	Frozen spec <code>docs/protocol-v1.md</code> with golden vectors in <code>fixtures/vectors/protocol-v1.json</code> ; where spec and code disagree the vectors decide. <code>/verify</code> re-checks the golden vectors in the browser on every load.

Windows claims rest on automated CI only; the scripted acts and chaos/tamper tools were run and evidenced on macOS.

Next

Field pilot on a real WAN with measured throughput and RTO; independent-custody witness under a separate organisation; TLS pinning between components; HSM-held sign-off key; real archive WORM storage; Tamil item bank with native review.

Links

- Repository: github.com/anirudh12032008/saakshi
- README: [README.md](#) (architecture, quickstart, how to run acts 1 to 5)
- Also in the repo: `docs/claims-ledger.md` , `docs/traceability.md` , `docs/threat-model.md` , `docs/protocol-v1.md`
- Demo video: submitted separately.

Licence: Apache-2.0.